



Answers & Solutions:

1. **B. maximum rate at which its output voltage can change.**

2. **D. push-pull amplifier.**

3. **B. 0.15 V**

Solution:

$$V_o = -\frac{1}{RC} \int V_i dt$$

$$V_o = -\frac{1}{(2.5 \times 10^{-6})(200 \times 10^3)} \int (-0.75) dt$$

at $t = 100$ ms

$$V_o = 0.15 \text{ V} \rightarrow \text{Ans.}$$

4. **A. pulse waveform.**

5. **C. 180° out of phase.**

6. **A. zero (0) V**

7. **A. differential amplifier**

8. **A. 12.0 V**

Solution:

$$V_o = A_o (V_2 - V_1) = 120(2.45 - 2.35)$$

$$V_o = 12 \text{ V} \rightarrow \text{Ans.}$$

9. **B. Differentiation**

10. **C. output offset voltage**

11. **B. a comparator with hysteresis**

12. **A. zero (0) V**

13. **D. the rate of change of the input voltage.**

14. **D. zero.**

15. **D. common-mode rejection ratio**

16. **C. a capacitor in its feedback circuit.**

17. **A. R_i .**

18. **C. 5 Hz**

Solution:

$$f_c = \frac{f_1}{A_{VD}} = \frac{1 \text{ MHz}}{200 \text{ V/mV}}$$

$$f_c = 5 \text{ Hz} \rightarrow \text{Ans.}$$

19. **B. a triangle wave**

20. **D. has the same potential as ground, yet it can sink no current.**

21. **D. 40**

Solution:

$$A_{CL} = \frac{\Delta V_o / \Delta t}{\Delta V_i / \Delta t}$$

$$A_{CL} = \frac{SR}{\Delta V_i / \Delta t} = \frac{2 \text{ V} / \mu\text{s}}{0.5 \text{ V} / \mu\text{s}}$$

$$A_{CL} = 40 \rightarrow \text{Ans.}$$

22. **C. Input offset voltage**

23. **B. +VSAT to -VSAT**

24. **C. Both of the above**

25. **B. 200,000.**

26. **D. 4.74**

Solution:

$$CMRR = 20 \log \left(\frac{A_d}{A_c} \right)$$

$$90 = 20 \log \left(\frac{150000}{A_c} \right)$$

$$A_c = 4.74 \rightarrow \text{Ans.}$$

27. **C. input offset current drift**

28. **A. very high input resistance**

29. **D. common gain common gain**

30. **B. roll-off**

31. **C. complementary amplifier**

32. **A. differential amplifier**

33. **B. zero**

34. **D. All of the above.**

35. **B. differential gain**

36. **B. 45.8mV**

Solution:

$$V_o = A_d V_d \left(1 + \frac{1}{CMRR} - \frac{V_c}{V_d} \right)$$

$$V_o = (4000)(10 \mu\text{V}) \left(1 + \frac{1}{100} - \frac{145 \mu\text{V}}{10 \mu\text{V}} \right)$$

$$V_o = 45.8 \text{ mV} \rightarrow \text{Ans.}$$

37. **C. virtual ground.**

38. **A. negative.**

39. **D. All of these are correct**

40. **B. a differentiator.**

41. **A. $\frac{1}{2} V_{cc}$.**

42. **A. -0.80 V**

Solution:

$$V_o = \left(-\frac{R_f}{R_i} \right) V_i$$

$$V_o = \left(-\frac{2000}{1000} \right) (0.4)$$

$$V_o = -0.8 \text{ V} \rightarrow \text{Ans.}$$

43. **D. +2.4 V**

Solution:

$$V_o = \left(-\frac{R_f}{R_i} \right) V_i$$

$$V_o = \left(-\frac{2000}{1000} \right) (-1.2)$$

$$V_o = +2.4 \text{ V} \rightarrow \text{Ans.}$$

44. **A. input bias current**

45. **B. break frequency**

46. **D. input offset current**

47. **D. common mode**



48. **B. op-amp**

49. **B. Schmitt trigger.**

50. **C. 0.99 mV**

Solution:

$$V_{os} = I_B \frac{R_i R_f}{R_i + R_f}$$

$$V_{os} = (100 \times 10^{-9}) \left[\frac{(10 \times 10^3)(1 \times 10^6)}{10 \times 10^3 + 1 \times 10^6} \right]$$

$$V_{os} = 0.99 \text{ mV} \rightarrow \text{Ans.}$$

END